

KiCad: An Electronics Schematic Design And Layout Creation Suite

Electronics design engineers require a number of software to create and verify their designs digitally, before moving to production. KiCad brings all these tools together under one platform, making their tasks simpler. This free electronic design automation (EDA) software suite mainly comprises a schematic creation and capture suite, and a PCB layout tool. It enables operations like creating bills of materials (BOMs), gerber files and PCB 3D views. Running under GPL v3 licence, this open source suite runs on Windows, Linux and MacOS.

Components of KiCad

KiCad suite integrates a number of popular open source tools for various EDA functions. The process incorporates the following steps: schematic creation, netlist generation, layout building and simulation. Given below are the tools that majorly make KiCad suite.

Eeschema. It is an interactive schematic creation platform that comes in KiCad suite. Eeschema brings a graphics-based design area that can create schematics with simple click operations. KiCad library is extensive and brings symbols, footprints and 3D models of almost all major components that are required for the circuit. This allows you to build any scale of circuit schematic on Eeschema.

Complex schematics can be broken into sub-circuits or into hierarchical circuitries. Many quick access buttons for components enable swift schematic creation.

Eeschema also supports im-

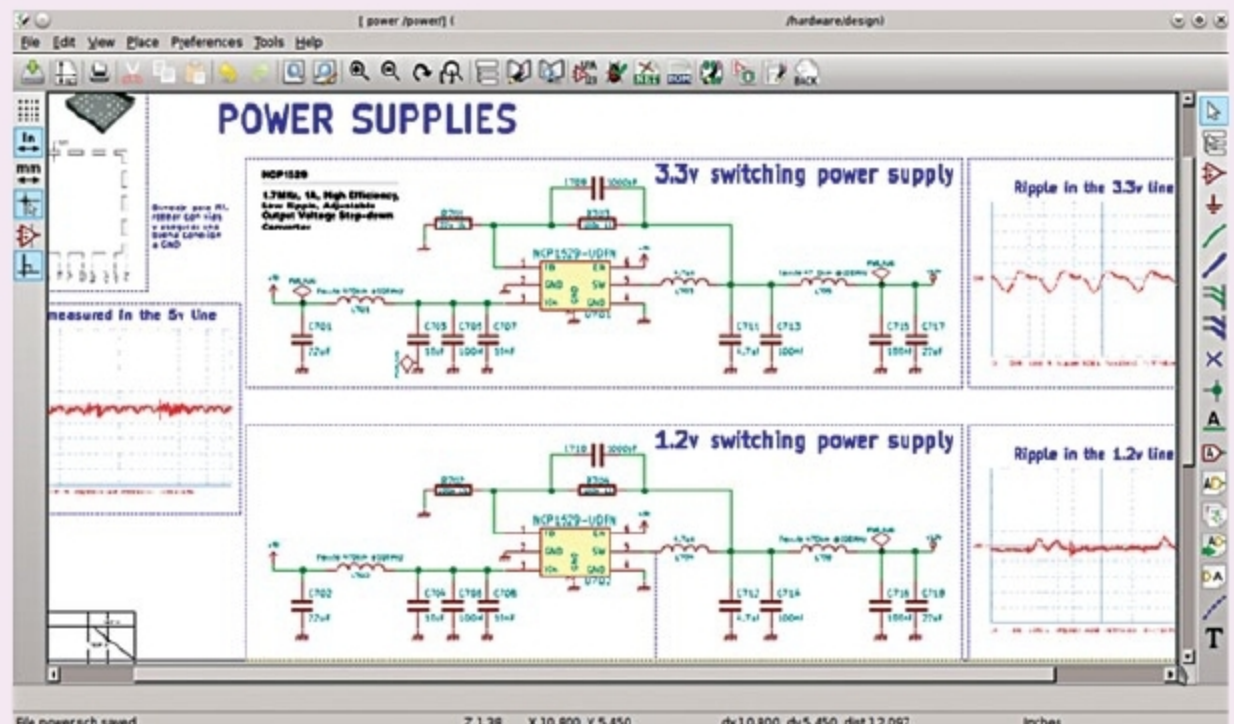


Fig. 1: A multi-level schematic on Eeschema tool, with text labelling (Credit: Wikipedia)

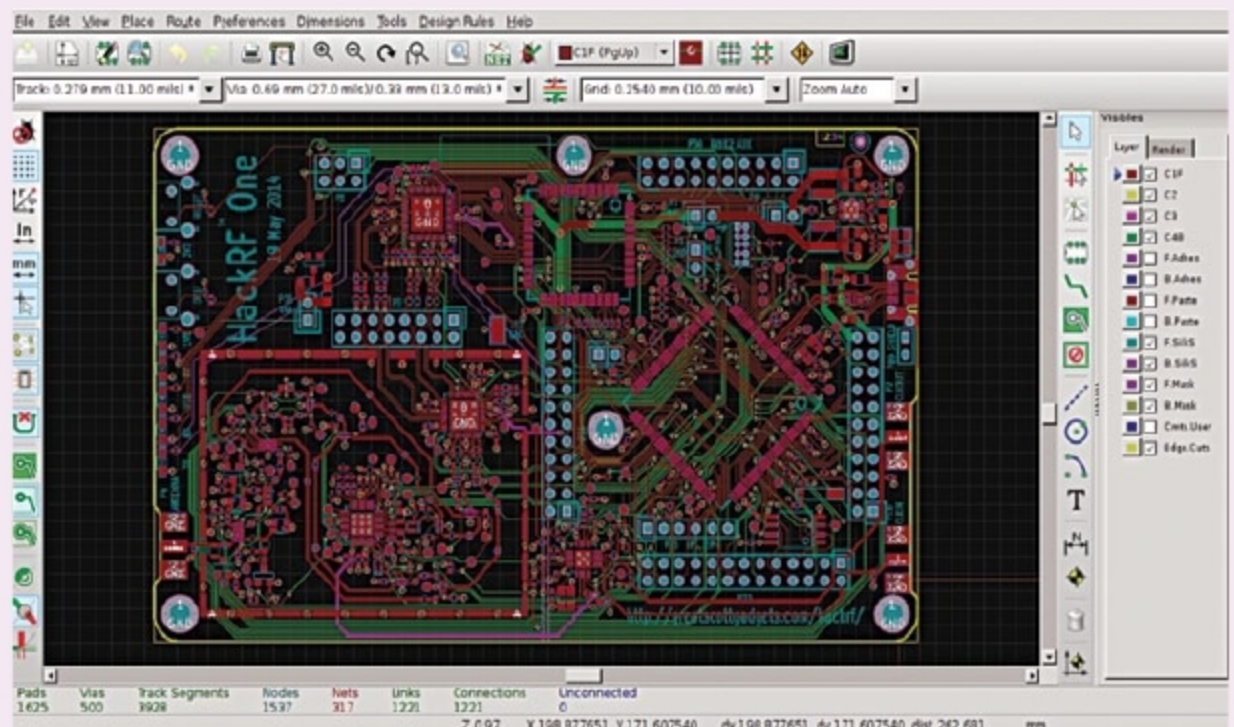


Fig. 2: PcbNew interface (Credit: kicad-pcb.org)

ported components library. This can be accessed from Preference menu.

Eeschema helps you verify your circuits through electrical rules check (ERC). It can point any conflicts in the output pin, unconnected pins or missing drivers. The schematic can be captured in a variety of formats, including netlists, which can be ex-

ported in various formats like Pspice, Cadstar, PcbNew and Generic XML. BOMs can also be generated using Eeschema. Buttons for both BOM and netlist creation are available on Eeschema's toolbar.

PcbNew. This tool is used to design the layout based on the schematic and netlist generated.